Introduction

Proteins are the molecular workhorses of life, orchestrating virtually every biological process from catalyzing reactions to transmitting signals and maintaining cellular structure. Understanding protein function from amino acid sequence alone remains one of the most fundamental challenges in biology. This challenge has direct implications across medicine, industry, and basic research, yet the vast majority of known proteins remain functionally uncharacterized.

Applications

Understanding protein function has profound implications across virtually every domain of biological research and biotechnology.

Drug Discovery
- Identifying therapeutic targets and predicting off-target effects by understanding which proteins participate in disease-relevant pathways

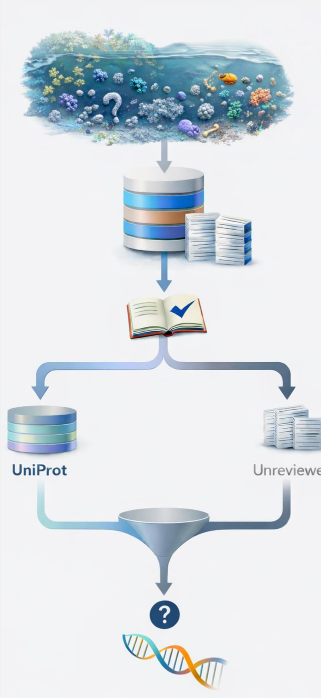
Precision Medicine
- Interpreting variants of unknown significance by predicting whether mutations disrupt protein function in patient genomes

Enzyme Engineering
- Screening metagenomic databases to identify promising biocatalysts for industrial applications including plastic degradation, carbon capture, and sustainable biofuel

Functional Genomics
- Annotating novel protein sequences from microbiome studies to reveal ecological roles and therapeutic potential hidden in environmental samples

The Dark Proteome

Despite the critical importance of understanding protein function, a massive gap exists between available sequence data and functional knowledge—creating a fundamental bottleneck limiting human progress against disease, environmental challenges, and industrial applications.



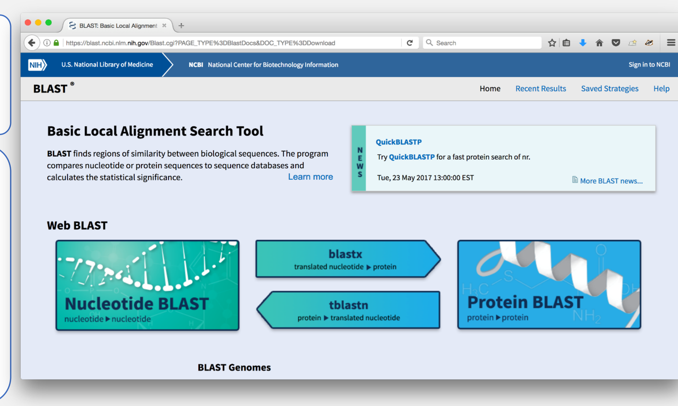
- High-throughput sequencing has cataloged ~246 million protein sequences in UniProt as of 2024
- Only ~574,000 entries (<0.25%) have experimentally validated functions through manual curation in Swiss-Prot
- The remaining 99.75% constitute the "dark proteome"—molecular machines whose therapeutic potential, ecological roles, and disease associations remain invisible to science
- The number of uncharacterized proteins continues to grow, adding approximately three hundred new sequences per minute
- Experimental characterization proceeds far too slowly to close this gap: a single protein's function can take weeks to months of laboratory work to determine

Traditional Computational Approaches

Consequently, the field has turned to computational prediction as the only scalable solution to bridge this widening gap between sequence availability and functional understanding.

BLAST: For over three decades, the most widely used tool for protein annotation; identifies sequences similar to already-characterized proteins and transfers their functional labels via homology-based inference

Limitations of homology transfer:
- Functional annotation is typically shallow even when similarity is plausible
- Can only annotate proteins that closely resemble already-characterized sequences
- Leaves the vast majority of the dark proteome—especially proteins from understudied organisms, extreme environments, and metagenomic discoveries—functionally invisible



Current Deep Learning Methods

More recently, deep learning methods have shown promise by learning latent relationships between sequences and functions that transcend simple sequence similarity, yet these approaches face their own critical limitations. Three key limitations persist.

Underutilization of protein language models (PLMs)
- Prevailing systems compress rich embeddings through aggressive low-dimensional bottlenecks, rely on a single PLM despite complementary inductive biases across architectures, or employ prediction frameworks that fail to fully express PLM representational capacity

Misattribution of performance drivers
- The field frequently overemphasizes structural inputs and places undue weight on auxiliary evolutionary information such as multiple sequence alignments (MSAs)

Auxiliary feature dependencies
- Requirements for predicted structures from computationally expensive pipelines (e.g., AlphaFold) or evolutionary alignments from database searches add substantial computational overhead, increase inference latency, limit applicability to novel proteins lacking evolutionary history, and risk propagating errors from upstream predictions

Research Objectives

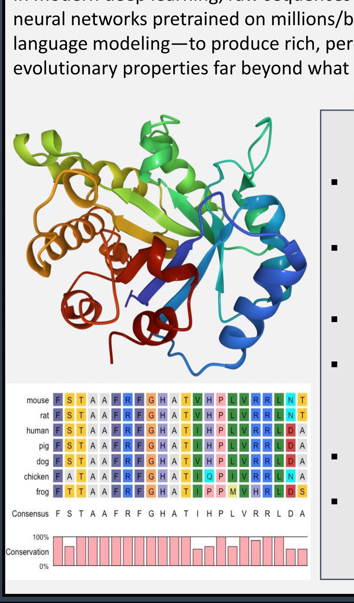
- Investigate the true drivers of performance gains in protein function prediction through systematic ablation experiments—isolating the contributions of auxiliary inputs (3D coordinates, MSAs) versus protein language model representations
- Design a PLM-centric architecture that fully leverages complementary protein language models through advanced fusion techniques (cross-attention, adaptive gating) and training strategies validated through systematic experimentation
- Evaluate rigorously on Gene Ontology prediction across all three ontologies (MF, BP, CC) against state-of-the-art methods, with additional benchmarking on enzyme classification and stability regression to assess generalization and fine-tuning capability
- Develop an accessible, serverless web application with a user-friendly interface for sequence-to-function prediction, alongside open-source implementation enabling mass annotation at scale

Protein Data Modality

Proteins are linear polymers defined entirely by their amino acid sequence—a chain of residues drawn from a 20-letter alphabet that encodes all the information necessary to determine a protein's structure, function, and behavior. This primary sequence is the most fundamental and universally available representation of any protein.

VRDAYIAKPENCVYECGITQDCNKLCTENGAESGYCQWGGKYGNACWCIKLPDSVPIRVPGKCQ

In modern deep learning, raw sequences are tokenized and passed through protein language models (PLMs)—large neural networks pretrained on millions/billions of protein sequences via self-supervised objectives such as masked language modeling—to produce rich, per-residue feature vectors (embeddings) that capture latent biochemical and evolutionary properties far beyond what raw one-hot encodings provide.



Auxiliary Information

3D Structure (Coordinates)
- Experimentally determined (e.g., Protein Data Bank) or computationally predicted (e.g., AlphaFold) three-dimensional atomic coordinates, represented as Cα backbone positions
- Often encoded as residue contact graphs or geometric features, providing explicit spatial proximity and geometric constraint information

Multiple Sequence Alignments (MSAs)
- Collections of evolutionarily related sequences aligned position-by-position, generated by searching large sequence databases for homologs
- Encode co-evolutionary information: correlated mutations across aligned positions reveal residue-residue contacts and functional constraints shaped by natural selection

Additional Modalities
- Other modalities include InterPro domain annotations, co-evolutionary coupling matrices, protein–protein interaction networks, and secondary structure
- These are not explicitly ablated in this work, as 3D coordinates and MSAs subsume much of the signal these sources provide, making them the most informative test cases for evaluating auxiliary feature dependence

Datasets & Evaluation Metrics

Gene Ontology

Task Background
- The Gene Ontology (GO) provides a controlled vocabulary for describing protein function across three orthogonal ontologies: molecular function (MF), biological process (BP), and cellular component (CC)
- MF captures specific biochemical activities (e.g., ATP binding, oxidoreductase activity); BP describes broader pathways (e.g., DNA repair, signal transduction); CC localizes proteins to subcellular structures (e.g., cytosol, ribosome)
- Technically challenging: labels are hierarchical, exhibit long-tailed frequency distributions, and require models to generalize to remote homologs and rare terms while remaining ontology-consistent
- Central task targeted in this work—regarded as one of the most important benchmarks in protein informatics since it is well-defined while addressing the key overall aspects of protein function through its three ontologies

Dataset
- PDBch: 36,641 protein chains from the Protein Data Bank clustered at 95% sequence identity, split 8:1:1 into 29,313 training, 3,664 validation, and 3,664 test proteins
- Label space: 489 MF terms, 1,943 BP terms, 320 CC terms—exhibiting moderate label imbalance typical of real-world annotation databases (rare terms ~50 examples, common terms in thousands)
- Test set remains fixed across all three ontologies for standard cross-study comparability

Evaluation Metrics
- GO predictions undergo standard hierarchy-aware propagation preprocessing to ensure consistency with the GO ontology tree
- **Protein-centric F_{max}:** Best thresholded trade-off between per-protein precision and recall; higher means balanced, reliable hits at the protein level
- **Function-centric $AUPR_{macro}$:** Averages average-precision over GO terms so rare and common labels count equally; higher means better performance on long-tailed terms
- **Semantic distance S_{min}:** Minimizes information-weighted error on the GO DAG; lower values indicate more semantically accurate and hierarchy-consistent predictions

$F_{\max}$	$\mathrm{AUPR}_{\mathrm{macro}}$	$S_{\min}$
$F_{\max} = \max_{\tau \in \mathcal{T}} \frac{2\,\overline{P}(\tau)\,\overline{R}(\tau)}{\overline{P}(\tau) + \overline{R}(\tau)}$	$\mathrm{AUPR}_{\mathrm{macro}} = \frac{1}{\lvert\mathcal{L}^{+}\rvert} \sum_{\ell \in \mathcal{L}^{+}} \mathrm{AP}_{\ell}$	$S_{\min} = \min_{\tau} \sqrt{\mathrm{RU}(\tau)^2 + \mathrm{MI}(\tau)^2}$

Enzyme Commission

Task Background
- EC number: hierarchical four-level taxonomy specifying the chemical transformation catalyzed by an enzyme (class → subclass → sub-subclass → substrate specificity)
- Near single-label classification with extremely long-tailed frequency distribution

Dataset
- Training: 74,487 sequences clustered at 70% identity from UniProt; Validation: NEW-392 (392 Swiss-Prot enzymes); Test: Price-149 (149 experimentally validated enzymes)

Evaluation Metric
- **AUC:** Area under ROC curve measuring class separability; higher means stronger discrimination

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR}), \quad \mathrm{TPR} = \frac{TP}{TP + FN}, \quad \mathrm{FPR} = \frac{FP}{FP + TN}.$$

Stability

Task Background
- Protein stability: ability of a sequence to retain its native fold under stress (heat, denaturants, proteolysis), summarized as a continuous score
- Regression task probing mutational landscape, models must generalize beyond single-mutation trends

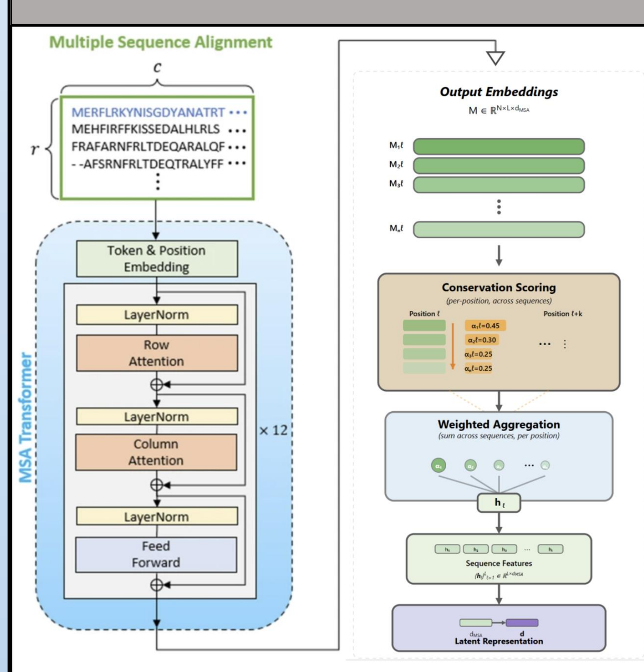
Dataset
- TAPE stability benchmark: 53,416 training, 2,512 validation, 12,851 test sequences from a high-throughput de novo miniprotein design campaign

Evaluation Metric
- **Spearman's ρ:** Rank correlation between predicted and true stability scores; higher means stronger monotonic agreement

$$\rho = \frac{\mathrm{Cov}(r(\mathbf{s}), r(\mathbf{y}))}{\sigma_{r(\mathbf{s})}\sigma_{r(\mathbf{y})}} = 1 - \frac{6\sum_{i=1}^{n}(r(s_i) - r(y_i))^2}{n(n^2 - 1)}.$$

Input Modality Investigation

Multiple Sequence Alignments



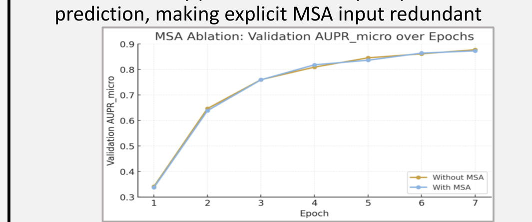
Background & Motivation
- Multiple sequence alignments (MSAs) capture evolutionary relationships by aligning homologous sequences, revealing conservation patterns and co-evolutionary signals
- Key question: Do MSAs provide measurable benefit to protein function prediction tasks when strong protein language models are available?

Data Generation
- Retrieved pre-computed MSAs from OpenFold dataset (BFD+UniClust30, MGnify, UniRef90) with Hamming diversity-maximizing subsampling
- For proteins absent from OpenFold: generated MSAs using ColabFold's MMseqs2 protocol (90% identity threshold, 75% coverage, max 256 sequences)

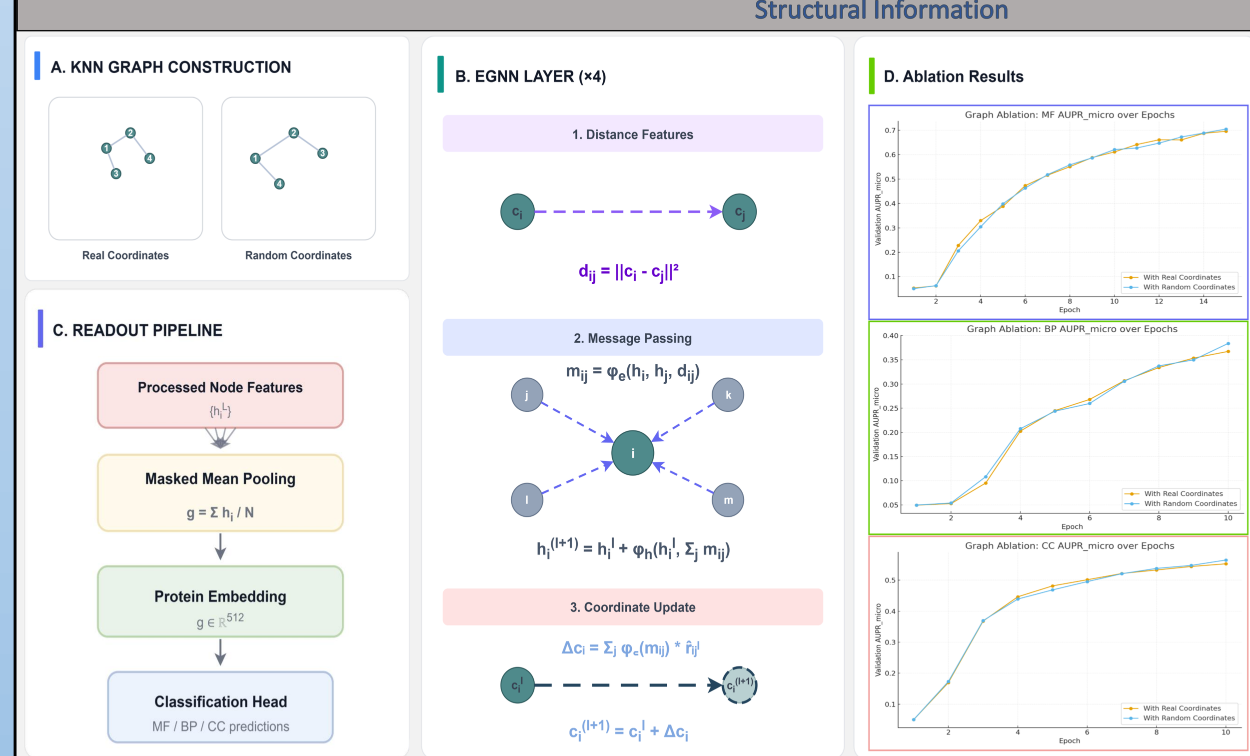
Experimental Setup
- Compared two architecturally identical models on MF prediction with varied inputs:
  - **Baseline:** Four PLM embeddings (ESM-C, ProtT5, Ankh3-XL, PGLM)
  - **MSA-augmented:** Pooled MSA features replace PGLM as fourth stream
- Substitution design ensures identical capacity, isolating effect of auxiliary evolutionary information vs single-sequence modeling
- MSAs encoded via MSA Transformer with learned conservation-weighted pooling (see diagram):

$$s_{n\ell} = \mathbf{w}^{\top}\mathbf{M}_{n\ell}, \qquad \alpha_{n\ell} = \frac{\exp(s_{n\ell}/(\tau\sqrt{N}))}{\sum_{n'} \exp(s_{n'\ell}/(\tau\sqrt{N}))}, \qquad \mathbf{h}_{\ell} = \sum_{n=1}^{N} \alpha_{n\ell}\mathbf{M}_{n\ell},$$

Results & Interpretation
- Final validation $AUPR_{micro}$: 0.877 (without MSA) vs 0.872 (with MSA); MSA features provide minimal measurable benefit when protein language models exist as a prior
- Interpretation: Modern PLMs pretrained on millions/billions of sequences have internalized relevant co-evolutionary patterns necessary for protein function prediction, making explicit MSA input redundant



Structural Information



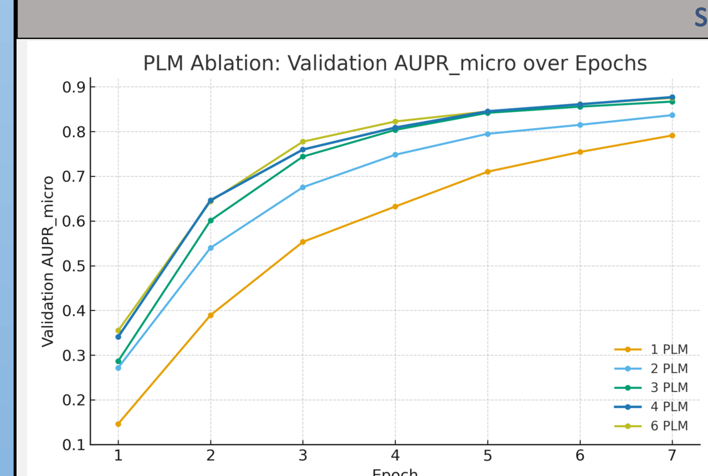
Background & Motivation
- 3D coordinates/contact maps traditionally considered essential for capturing geometric constraints and spatial relationships between residues
- Key question: Does structural geometry provide measurable benefit when strong PLM representations are available?

Data Generation
- **Real coordinates:** Experimentally determined Cα positions from Protein Data Bank (PDB) structures
- **Random coordinates:** i.i.d. random point clouds of same length (no structural signal), k-NN graph constructed identically

Experimental Setup
- Trained E(n)-equivariant graph neural network (EGNN) on all three gene ontologies with described varied inputs, comparing real and random coordinates (see diagram)
- Nodes initialized with ESM-C PLM embeddings to purely isolate the impact of the geometry itself

Results & Interpretation
- MF: 0.696 real vs 0.705 random | BP: 0.367 real vs 0.374 random | CC: 0.553 real vs 0.565 random
- Random coordinates achieve nearly identical performance across all ontologies
- Interpretation: PLMs have internalized high-level spatial proximity patterns through co-occurrence statistics for protein function prediction—explicit 3D geometry provides negligible additional signal

Scaling Protein Language Models



Background & Motivation
- Nearly all existing prediction methods rely on a single PLM to extract sequence features
- Key question: Does incorporating multiple PLMs with diverse architectural biases improve performance?

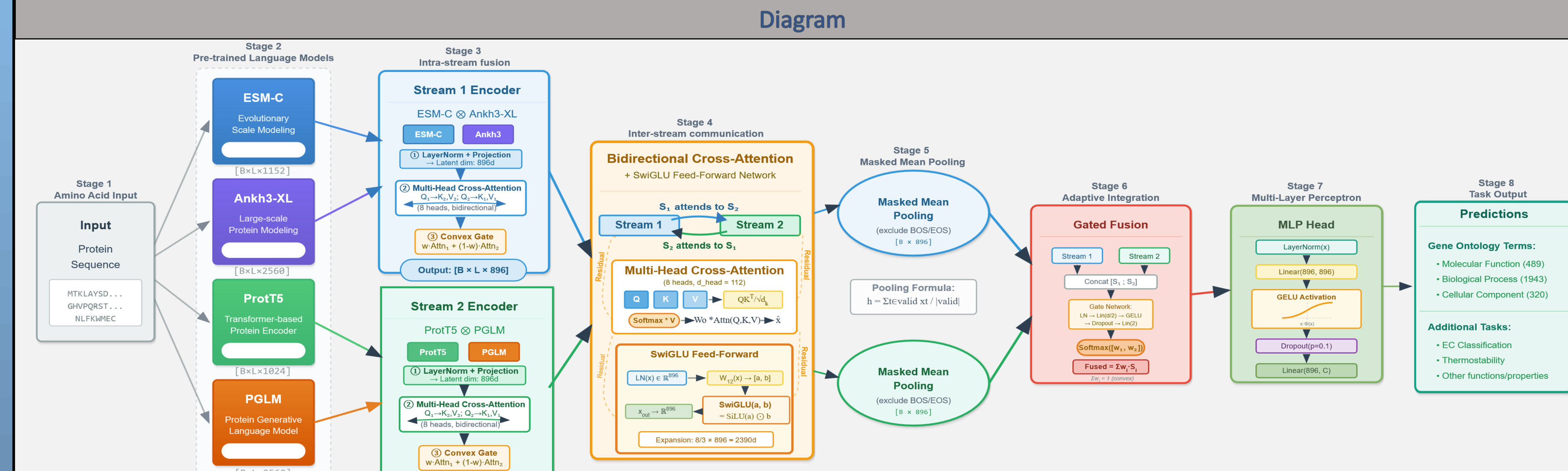
Epoch	1	2	3	4	6
1	0.146	0.272	0.287	0.341	0.355
2	0.389	0.540	0.601	0.646	0.644
3	0.553	0.675	0.744	0.760	0.777
4	0.632	0.745	0.804	0.809	0.822
5	0.710	0.795	0.842	0.845	0.845
6	0.754	0.815	0.856	0.861	0.861
7	**0.791**	**0.837**	**0.867**	**0.877**	**0.875**

Experimental Setup
- Systematically varied number of PLM streams on MF prediction over first 7 epochs
- Starting from ESM-C baseline, incrementally added: ProtT5 → Ankh3-XL → PGLM → (MSA Transformer + GENIE)
- Architecture adapts to PLM count: 1 PLM = self-attention encoder; 2 PLMs = bidirectional cross-attention; 3+ PLMs = dual-stream encoder with grouped representations
- All variants hold fixed modules to control for model capacity

Results & Interpretation
- Performance improves monotonically from 1→4 PLMs (epoch 7: 0.791 → 0.837 → 0.867 → 0.877)
- Beyond 4 PLMs, gains plateau (6 PLMs: 0.875)—diminishing returns from redundancy
- Interpretation: Each PLM contributes distinct inductive biases from its architecture, pretraining corpus, and training objective—complementary representations drive performance gains
- Optimal sweet spot: 3-4 complementary PLMs for protein representation learning

Architecture

Diagram



Overview
- Four complementary PLMs with distinct inductive biases: ESM-C (efficiency-scaled MLM), ProtT5 (encoder-decoder long-range dependencies), Ankh3-XL (multi-task pretraining), PGLM (unified autoencoding + autoregressive)
- Dual-Stream Encoder: PLMs grouped into two streams (ESM-C + Ankh3 | ProtT5 + PGLM)—intra-stream fusion via bidirectional cross-attention with convex gating produces per-stream representations
- Bidirectional Cross-Attention Transformer: Streams exchange information through N layers of symmetric cross-attention (each stream attends to the other) followed by SwiGLU feed-forward networks
- Masked Mean Pooling: Aggregates token-level representations to protein-level embeddings, excluding padding and BOS/EOS registers
- Convex Gated Fusion: Learned softmax-normalized weights (summing to 1) combine stream embeddings—ensuring stable mixing
- MLP Prediction Head: Task-specific output layer produces GO terms, EC numbers, stability scores, or any other relevant objective
- Key design principle: enable PLMs to communicate and share complementary insights via cross-attention—allowing selective querying of relevant features across streams—rather than relying on simple concatenation, while incorporating cutting-edge mathematical techniques and disregarding redundant auxiliary inputs

Training Techniques

Loss Function
- Asymmetric Loss (ASL) for multi-label classification (GO, EC)—focal-style objective addressing extreme class imbalance:

$$\mathcal{L}_{\mathrm{ASL}}(\mathbf{x}, \mathbf{y}) = -\sum_{c=1}^{C}\left[y_c \cdot \log(p_c^{+}) + (1 - y_c) \cdot \log(p_c^{-})\right] \cdot w_c,$$

$$w_c = (1 - p_c^{+} \cdot y_c - p_c^{-} \cdot (1 - y_c))^{\gamma_c}, \qquad \gamma_c = \gamma_{\mathrm{pos}} \cdot y_c + \gamma_{\mathrm{neg}} \cdot (1 - y_c),$$

- Standard Mean Squared Error (MSE) loss for stability regression:

$$\mathcal{L}_{\mathrm{MSE}}(\mathbf{x}, \mathbf{y}) = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2,$$

Learning Rate Scheduler
- Cosine annealing with linear warmup (5% of training steps)
- Warmup stabilizes early optimization; cosine decay ensures smooth convergence

$$\eta_t = \begin{cases} \eta \cdot \frac{t}{T_{\mathrm{warmup}}}, & \text{if } t < T_{\mathrm{warmup}} \quad \text{(linear warmup)}, \\ \eta \cdot \frac{1}{2}\left(1 + \cos\left(\pi \cdot \frac{t - T_{\mathrm{warmup}}}{T - T_{\mathrm{warmup}}}\right)\right), & \text{otherwise} \quad \text{(cosine decay)}. \end{cases}$$

Weights Averaging Scheme
- Exponential Moving Average (decay τ = 0.999), improves model robustness and generalization
- EMA accumulation begins after warmup; EMA weights used at inference

$$\theta_{\mathrm{EMA}}^{(t)} = \tau \cdot \theta_{\mathrm{EMA}}^{(t-1)} + (1 - \tau) \cdot \theta^{(t)}$$

Optimization
- AdamW for most efficient convergence and loss exploration
- Learning rate $\eta = 10^{-4}$ for parameters, weight decay $\lambda = 0.01$
- Momentum: $\beta_1 = 0.9$, $\beta_2 = 0.999$, numerical stability $\epsilon = 10^{-8}$
- Gradient clipping at norm 1.0 to prevent training instability

Early Stopping
- Monitors validation performance with patience of 2 epochs
- Minimum improvement threshold Δ_min = 0.003
- Tracks $\mathrm{AUPR}_{\mathrm{micro}}$ for classification tasks, Spearman's ρ in regression

Ablations

Epoch	1	2	3	4	5	6	7
Bidirectional Transformer vs Self-Attention Transformer							
Bidirectional Transformer	0.3372	0.6382	0.7592	0.8180	0.8358	0.8638	**0.8724**
Self-Attention Transformer	0.3318	0.6180	0.7288	0.7895	0.8129	0.8383	**0.8589**
Δ (A−B)	+0.0054	+0.0202	+0.0304	+0.0285	+0.0229	+0.0255	**+0.0135**
Cross Attention Pooling vs Naive Concatenation Pooling							
Cross Attention Pooling	0.2695	0.5756	0.7300	0.7841	0.8282	0.8490	**0.8649**
Naive Concatenation Pooling	0.2518	0.5364	0.6960	0.7764	0.8167	0.8505	**0.8521**
Δ (A−B)	+0.0177	+0.0392	+0.0340	+0.0078	+0.0114	-0.0015	**+0.0128**
SwiGLU + Register Tokens							
With SwiGLU + Registers	0.2866	0.6012	0.7439	0.8036	0.8418	0.8556	**0.8669**
No SwiGLU/Registers	0.2518	0.5364	0.6960	0.7764	0.8167	0.8505	**0.8521**
Δ (A−B)	+0.0348	+0.0649	+0.0479	+0.0273	+0.0250	+0.0051	**+0.0148**
Convex Gating vs Per-Dim Gating							
Convex Gating	0.2866	0.6012	0.7439	0.8036	0.8418	0.8556	**0.8669**
Per-Dim Gating	0.3143	0.6316	0.7581	0.8196	0.8358	0.8586	**0.8592**
Δ (A−B)	-0.0277	-0.0304	-0.0141	-0.0160	+0.0060	-0.0030	**+0.0076**
High-Dimensional Embedding vs Low-Dimensional Embedding							
High-Dimensional Embedding	0.3408	0.6464	0.7596	0.8088	0.8453	0.8609	**0.8770**
Low-Dimensional Embedding	0.2995	0.6329	0.7459	0.8069	0.8425	0.8599	**0.8724**
Δ (A−B)	+0.0413	+0.0135	+0.0137	+0.0019	+0.0028	+0.0010	**+0.0045**

Experiment Setup
- Systematic ablations on MF prediction task to validate architectural design choices
- All models trained for 7 epochs and evaluated using validation $\mathrm{AUPR}_{\mathrm{micro}}$
- Five architectural dimensions tested:
  - **Attention mechanism:** Bidirectional cross-attention between streams vs independent self-attention within each stream
  - **Pooling strategy:** Cross-attention streamed pooling vs naive concatenated projection pooling
  - **Architectural enhancements:** SwiGLU activations + learnable register tokens vs standard configurations
  - **Gating strategy:** Convex (softmax-normalized) vs per-dimension element-wise gating
  - **Embedding dimensionality:** High-dimensional (Ankh3-XL, 2560d) vs low-dimensional (Ankh3-L, 1536d)

- Collectively: Design choices are complementary and compound to form a well-calibrated architecture
- Additional experiments (omitted for brevity): loss formulation (ASL outperformed BCE for handling extreme class imbalance), optimal latent dimensionality (896 balanced capacity and regularization), weight-averaging strategy (EMA improved inference smoothness over standard weights), learning rate schedules (cosine annealing with warmup stabilized convergence), optimizer comparisons, etc.

Algorithm

Algorithm 1 Omni-Prot: PLM-Centric Fusion for Protein Function

Require: Amino acid sequence $\mathbf{s} = (s_1, \ldots, s_L)$
Ensure: Prediction vector $\hat{\mathbf{y}} \in \mathbb{R}^C$

Stage 1: PLM Feature Extraction
1: **for** $m \in \{\text{ESM-C}, \text{ProtT5}, \text{Ankh3-XL}, \text{PGLM}\}$ **do**
2: $\quad \mathbf{H}^{(m)} \leftarrow \mathcal{E}_m(\mathbf{s}) \in \mathbb{R}^{L \times d_m}$
3: **end for**

Stage 2: Projection to Shared Latent Space
4: **for** each PLM embedding $\mathbf{H}^{(m)}$ **do**
5: $\quad \mathbf{Z}^{(m)} \leftarrow \mathrm{Dropout}(\mathrm{Proj}(\mathrm{LN}(\mathbf{H}^{(m)}))) \in \mathbb{R}^{L \times d}$
6: $\quad$ Append [BOS]/[EOS] register tokens to $\mathbf{Z}^{(m)}$
7: **end for**

Stage 3: Dual-Stream Encoder (Intra-Stream Fusion)
8: **for** $(i, j) \in \{(\text{ESM-C}, \text{Ankh3}), (\text{ProtT5}, \text{PGLM})\}$ **do**
9: $\quad \tilde{\mathbf{Z}}^{(i)} \leftarrow \mathrm{MHA}(\mathrm{LN}(\mathbf{Z}^{(i)}), \mathrm{LN}(\mathbf{Z}^{(j)}), \mathrm{LN}(\mathbf{Z}^{(j)}))$
10: $\quad \tilde{\mathbf{Z}}^{(j)} \leftarrow \mathrm{MHA}(\mathrm{LN}(\mathbf{Z}^{(j)}), \mathrm{LN}(\mathbf{Z}^{(i)}), \mathrm{LN}(\mathbf{Z}^{(i)}))$
11: $\quad \mathbf{W}^{(i,j)} \leftarrow \mathrm{softmax}\left(\mathrm{MLP}_{\mathrm{gate}}(\tilde{\mathbf{Z}}^{(i)} \oplus \tilde{\mathbf{Z}}^{(j)})\right)$
12: $\quad \mathbf{F}^{(i,j)} \leftarrow \mathbf{W}^{(i,j)}_{:,1} \odot \tilde{\mathbf{Z}}^{(i)} + \mathbf{W}^{(i,j)}_{:,2} \odot \tilde{\mathbf{Z}}^{(j)}$
13: **end for**
14: $\mathbf{X}^{(0)} \leftarrow \mathbf{F}^{(\text{ESM-C},\text{Ankh3})}; \quad \mathbf{Y}^{(0)} \leftarrow \mathbf{F}^{(\text{ProtT5},\text{PGLM})}$

Stage 4: Bidirectional Cross-Attention Transformer
15: **for** $\ell = 1, \ldots, N$ **do**
16: $\quad \tilde{\mathbf{X}}^{(\ell)} \leftarrow \mathbf{X}^{(\ell-1)} + \mathrm{MHA}(\mathrm{LN}(\mathbf{X}^{(\ell-1)}), \mathrm{LN}(\mathbf{Y}^{(\ell-1)}), \mathrm{LN}(\mathbf{Y}^{(\ell-1)}))$
17: $\quad \tilde{\mathbf{Y}}^{(\ell)} \leftarrow \mathbf{Y}^{(\ell-1)} + \mathrm{MHA}(\mathrm{LN}(\mathbf{Y}^{(\ell-1)}), \mathrm{LN}(\mathbf{X}^{(\ell-1)}), \mathrm{LN}(\mathbf{X}^{(\ell-1)}))$
18: $\quad \mathbf{X}^{(\ell)} \leftarrow \tilde{\mathbf{X}}^{(\ell)} + \mathrm{FFN}_{\mathrm{SwiGLU}}(\mathrm{LN}(\tilde{\mathbf{X}}^{(\ell)}))$
19: $\quad \mathbf{Y}^{(\ell)} \leftarrow \tilde{\mathbf{Y}}^{(\ell)} + \mathrm{FFN}_{\mathrm{SwiGLU}}(\mathrm{LN}(\tilde{\mathbf{Y}}^{(\ell)}))$
20: **end for**

Stage 5: Masked Mean Pooling
21: $\mathbf{x}_{\mathrm{pool}} \leftarrow \frac{\sum_t M_t \mathbf{X}_t^{(N)}}{\sum_t M_t + \epsilon}; \quad \mathbf{y}_{\mathrm{pool}} \leftarrow \frac{\sum_t M_t \mathbf{Y}_t^{(N)}}{\sum_t M_t + \epsilon}$

Stage 6: Convex Gated Fusion
22: $[\alpha, \beta] \leftarrow \mathrm{softmax}\left(\mathrm{MLP}_{\mathrm{fuse}}(\mathbf{x}_{\mathrm{pool}} \oplus \mathbf{y}_{\mathrm{pool}})\right)$
23: $\mathbf{z} \leftarrow \alpha \cdot \mathbf{x}_{\mathrm{pool}} + \beta \cdot \mathbf{y}_{\mathrm{pool}}$

Stage 7: Prediction Head
24: $\hat{\mathbf{y}} \leftarrow \mathrm{MLP}_{\mathrm{head}}(\mathbf{z})$
25: **return** $\hat{\mathbf{y}}$

Multi-Head Attention:

$$\mathrm{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Concat}_{r=1}^{H}\left[\mathrm{Attn}(\mathbf{Q}\mathbf{W}_r^Q, \mathbf{K}\mathbf{W}_r^K, \mathbf{V}\mathbf{W}_r^V)\right]\mathbf{W}^O$$

Scaled Dot-Product Attention:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_{\mathrm{head}}}}\right)\mathbf{V}$$

SwiGLU Feed-Forward:

$$\mathrm{FFN}_{\mathrm{SwiGLU}}(\mathbf{h}) = \left(\mathrm{SiLU}(\mathbf{h}\mathbf{W}_1) \odot \mathbf{h}\mathbf{W}_2\right)\mathbf{W}_3$$

MLP block (used for gate/fuse/head):

$$\mathrm{MLP}_k(\mathbf{u}) = \mathbf{W}_2^{(k)}\,\mathrm{Dropout}\left(\mathrm{GELU}(\mathbf{W}_1^{(k)}\,\mathrm{LN}(\mathbf{u}))\right), \qquad k \in \{\text{gate}, \text{fuse}, \text{head}\}.$$

$$\mathrm{MLP}_{\mathrm{gate}}, \mathrm{MLP}_{\mathrm{fuse}} : \mathbb{R}^{2d} \to \mathbb{R}^2 \text{ (hidden } d/2), \qquad \mathrm{MLP}_{\mathrm{head}} : \mathbb{R}^d \to \mathbb{R}^C \text{ (hidden } d).$$

Architecture constants:

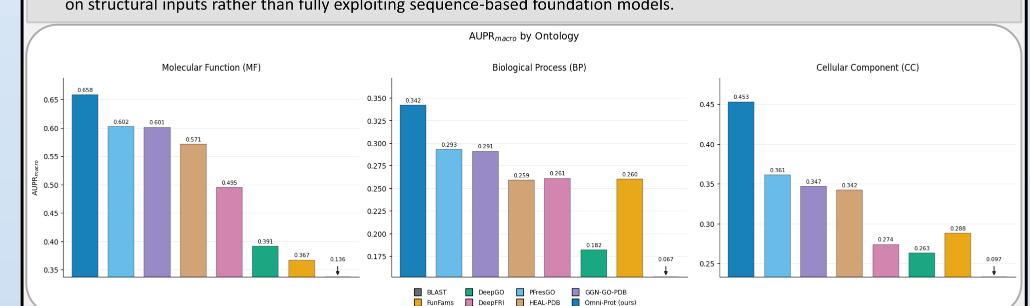
$$d = 896, \quad H = 8, \quad d_{\mathrm{head}} = 112, \quad d_{\mathrm{hidden}} \approx \frac{8d}{3}, \quad N = 3$$

Results

Gene Ontology

(Main Task)	Molecular Function (MF)			Biological Process (BP)			Cellular Component (CC)		
Method	$\mathrm{AUPR}_{\mathrm{macro}}$	$F_{\max}$	$S_{\min}$	$\mathrm{AUPR}_{\mathrm{macro}}$	$F_{\max}$	$S_{\min}$	$\mathrm{AUPR}_{\mathrm{macro}}$	$F_{\max}$	$S_{\min}$
BLAST	0.136	0.328	0.632	0.067	0.336	0.651	0.097	0.448	0.628
FunFams	0.367	0.572	0.531	0.260	0.500	0.579	0.288	0.672	0.503
DeepGO	0.391	0.577	0.472	0.182	0.493	0.577	0.263	0.549	0.550
DeepFRI	0.495	0.625	0.437	0.261	0.540	0.543	0.274	0.613	0.527
PFresGO	0.602	0.692	0.417	0.293	0.568	0.535	0.361	0.674	0.498
HEAL-PDB	0.571	0.691	0.401	0.259	0.565	0.540	0.342	0.655	0.501
GGN-GO-PDB	0.601	0.698	0.400	0.291	**0.643**	0.546	0.347	0.657	0.510
Omni-Prot (ours)	**0.658**	**0.743**	**0.342**	**0.342**	0.615	**0.498**	**0.453**	**0.728**	**0.420**

- **BLAST:** Transfers protein functions via nearest-neighbor local sequence alignment; commonly used as a classical homology-transfer baseline.
- **FunFams:** Subdivides CATH homologous superfamilies into functionally coherent FunFams for domain-level function annotation/transfer.
- **DeepGO:** An early deep-learning approach that learns sequence and cross-species protein–protein interaction features while explicitly modeling Gene Ontology hierarchical dependencies.
- **DeepFRI:** Combines protein language model sequence embeddings with graph neural networks over residue–residue contact maps to predict GO terms.
- **PFresGO:** An attention-based framework that integrates Gene Ontology graph relationships with representations extracted from protein language models.
- **HEAL:** Employs a hierarchical graph transformer trained with contrastive learning on structural graphs, using protein language model embeddings as node features for function prediction.
- **GGN-GO:** A geometric graph neural network leveraging protein language model embeddings and multiscale structural representations (atomic and residue levels) through geometric vector perceptrons; the prior state-of-the-art, with heavy reliance on structural inputs rather than fully exploiting sequence-based foundation models.



- State-of-the-art across MF and CC on all metrics; BP shows strong ranking improvement ($\mathrm{AUPR}_{\mathrm{macro}}$) with competitive $F_{\max}$
- Molecular Function: $\mathrm{AUPR}_{\mathrm{macro}}$ +9.35% (0.602→0.658), $F_{\max}$ +6.50% (0.698→0.743), $S_{\min}$ −14.4% (0.400→0.342)—gains in MF directly translate to more accurate identification of specific enzymatic activities and binding capabilities, beneficial for drug target identification
- Biological Process: $\mathrm{AUPR}_{\mathrm{macro}}$ +16.7% (0.293→0.342), $S_{\min}$ −7.0% (0.535→0.498)—substantial $\mathrm{AUPR}_{\mathrm{macro}}$ improvement indicates markedly better ranking of rare biological processes, suggesting the model captures complex pathway-level relationships more effectively despite BP's inherent difficulty in threshold-dependent metrics
- Cellular Component: $\mathrm{AUPR}_{\mathrm{macro}}$ +25.4% (0.361→0.453), $F_{\max}$ +8.05% (0.674→0.728), $S_{\min}$ −15.7% (0.498→0.420)—dramatic $\mathrm{AUPR}_{\mathrm{macro}}$ gains reflect substantially improved discrimination of rare subcellular compartments and protein complexes, particularly valuable for understanding cellular organization in understudied organisms
- Consistent $\mathrm{AUPR}_{\mathrm{macro}}$ improvements across all ontologies demonstrate superior handling of long-tailed label distributions—critical for real-world annotation where rare terms are often most biologically interesting
- Achieved using sequence-only input—outperforms methods relying on structural geometry or evolutionary alignments, validating PLM-centric design

Enzyme Commission

(Supplemental Task)

Method	AUC
GrAPFI	0.5095
ECPICK	0.5888
CLEAN	0.7334
GraphEC	0.8404
Omni-Prot (ours)	**0.855**

Gains
Against the strongest prior, GraphEC (0.8404), Omni-Prot attains an AUC score of 0.855 (Δ = +0.0146; +1.74%). Gains are larger versus earlier baselines: CLEAN (0.7334) → Δ = +0.1216; +16.58%, ECPICK (0.5888) → Δ = +0.2662; +45.21%, and GrAPFI (0.5095) → Δ = +0.3455; +67.81%.

Prior Methods
- **GrAPFI:** Constructs a weighted protein graph from InterPro domain-composition similarity and propagates EC labels from neighboring proteins.
- **ECPICK:** Uses evidential deep learning to generate EC predictions with calibrated uncertainty, improving reliability and trustworthiness on large-scale enzyme databases.
- **CLEAN:** Applies contrastive learning to protein language model representations using curated positive and negative pairs, enabling accurate EC annotation and identification of mislabeled or promiscuous enzymes.
- **GraphEC:** A geometric graph-based framework that combines ESMFold-predicted structures with protein language model embeddings, localizes active sites, and refines EC predictions through label diffusion.

Stability

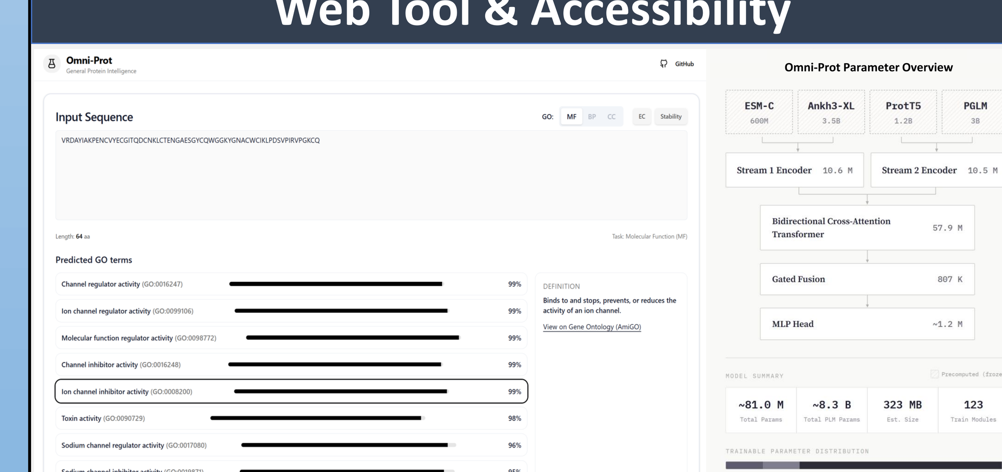
(Supplemental Task)

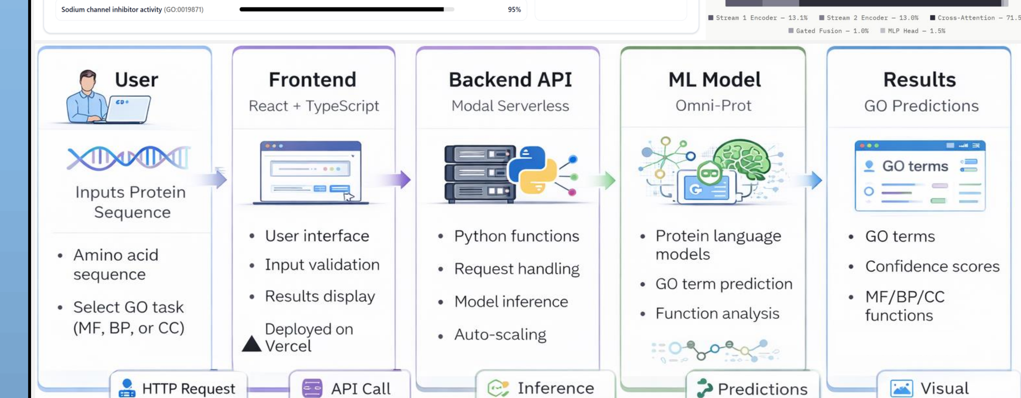
Method	Spearman ρ
One-Hot	0.19
k-NN	0.21
LSTM	0.69
RT ($\mathcal{L}_{SC}$)	0.71
Transformer	0.73
UniRep	0.73
ProteinBERT	0.76
Omni-Prot (ours)	**0.85**

Gains
Relative to the strongest prior, ProteinBERT (0.76), Omni-Prot reaches a ρ of 0.85 (Δ = +0.09; +11.8%). Gains vs other widely used baselines are consistent: Transformer (0.73) → Δ = +0.12; +16.4%, LSTM (0.69) → Δ = +0.16; +23.2%, RTLSC (0.71) → Δ = +0.14; +19.7%, k-NN (0.21) → Δ = +0.64; +305%, and One-Hot (0.19) → Δ = +0.66; +347%.

Prior Methods
- **k-NN:** A non-parametric nearest-neighbor regression approach based on sequence-similarity features, using Levenshtein distance between protein sequences.
- **One-Hot MLP:** A multilayer perceptron trained over one-hot residue encodings for stability prediction.
- **LSTM:** A self-supervised pretrained bidirectional LSTM sequence model fine-tuned for stability regression.
- **Regression Transformer (LSC):** An XLNet-based multitask sequence model using a self-consistency loss for protein property regression and conditional sequence generation.
- **Transformer:** A pretrained self-attention encoder fine-tuned for stability regression.
- **UniRep:** A multiplicative LSTM pretrained on large unlabeled protein sequence corpora and subsequently fine-tuned for stability prediction tasks.
- **ProteinBERT:** A protein-specific BERT-style model pretrained on large-scale sequence data and adapted with supervised heads for downstream stability prediction.

Web Tool & Accessibility





Web Application Release & Motivation
- Released user-friendly web application for immediate practical use, designed primarily for GO term prediction where the approach shows strongest performance improvements
- Many existing deep learning models for protein function prediction lack accessible web interfaces for practical deployment, limiting use by broader research community
- Current web-based protein function tools rely on BLAST and homology transfer, which cannot learn functional patterns beyond sequence similarity to close database matches

Performance & Applications
- Model overcomes these limitations while providing massive improvements compared to existing homology-based methods
- Particularly useful for annotating metagenomic discoveries, interpreting disease variants, and accelerating pathway design (within supported GO dictionary)
- Tool optionally supports EC classification and stability prediction for enzyme engineering applications

Key Advantages
- Enables scalable mass annotation of proteins for large-scale analyses due to model's sole reliance on amino acid sequences—a critical capability for high-throughput proteomics and functional genomics studies
- Modular architecture enables researchers to adapt framework to new tasks through fine-tuning on custom datasets
- Researchers can seamlessly integrate more powerful PLMs as they emerge, ensuring long-term relevance without architectural redesign

Discussion

Future Impact
- Drug discovery: GO predictions for molecular function and biological process directly inform target prioritization—identifying which uncharacterized proteins participate in disease-relevant pathways while flagging potential off-target effects through predicted functional similarities
- Precision medicine: Cellular component and molecular function predictions provide interpretive context for variants of unknown significance—enabling clinicians to assess whether mutations likely disrupt protein function or localization
- Enzyme engineering: Sequence-only inference enables rapid screening of massive metagenomic databases for promising biocatalysts when evaluating candidates for plastic degradation, carbon capture, and biofuel production
- Metagenomics: Massive improvements over BLAST enable functional annotation of novel sequences from understudied organisms and extreme environments—proteins that homology-based methods leave functionally invisible
- Scalable deployment: Independence from MSA generation and structure prediction enables true high-throughput analysis—annotating thousands of proteins simultaneously without computational bottlenecks
- Extensibility: Modular architecture allows seamless integration of more powerful PLMs as they emerge—ensuring long-term relevance without architectural redesign or retraining from scratch

Limitations
- Computational constraints prevented extensive repeated trials across all experimental configurations—though preliminary experiments showed low variance
- EC dataset exhibits label inconsistencies between training and test splits and extreme label imbalance—may affect absolute performance metrics for the secondary task
- Performance on proteins from deeply understudied taxonomic lineages, extreme environments, or synthetic designs diverging substantially from natural evolutionary trajectories may vary
- Future work should systematically evaluate robustness across underrepresented protein spaces

Conclusion
- Omni-Prot achieves state-of-the-art on GO prediction (all three ontologies), EC classification, and stability regression using sequence-only input
- Auxiliary inputs (3D coordinates, MSAs) provide negligible benefit when strong PLM representations are properly leveraged
- Future research should focus on extracting information and increasing capable performance of protein language models to maximize performance on a wide array of protein function prediction tasks
- Multi-PLM fusion with complementary architectural biases drives performance—validated through systematic ablation
- Sequence-only design eliminates error propagation from upstream pipelines, simplifies deployment, and enables greater potential robustness in terms of generalization to proteins lacking evolutionary history
- Open-source implementation and accessible web interface make advances immediately available to the broader research community